Software Requirements Specification

(SRS)

1. Introduction

1.1 Purpose

This document defines the complete software requirements for the AI Document Intelligence Platform. It serves as the formal agreement between stakeholders and the development team on what the system must do.

1.2 Scope

The platform is an internal company tool that provides three AI-powered document modules:

- **Case Study RAG:** Semantic search across past case studies to find relevant ones for new projects
- **Tender Analyzer:** Automated analysis of government tenders, requirement extraction, document matching, and folder assembly
- **Methodology Finder:** Quick search and retrieval of methodology documents

All modules are accessed through a single web application.

1.3 User Roles

Role	Description
Admin	Full access. Can manage users, upload/delete documents across all modules, configure system settings, and view audit logs.
User	Can use all three modules for searching and viewing. Can upload tenders for analysis. Cannot delete documents or manage other users.

2. Functional Requirements

2.1 Authentication & User Management

ID	Requirement
1.	The system shall allow users to register with email, name, and password.
2.	The system shall authenticate users via email and password, returning a JWT token.
3.	JWT tokens shall expire after 24 hours and support refresh.
4.	The system shall enforce role-based access control (Admin, User) on all endpoints.
5.	Admin users shall be able to create, edit, deactivate, and delete user accounts.
6.	Passwords shall be hashed using bcrypt before storage.
7.	The system shall lock an account after 5 consecutive failed login attempts for 15 minutes.

2.2 Case Study Module

2.2.1 Document Ingestion

ID	Requirement
1.	The system shall accept case study uploads in PDF, DOCX, PPTX, and image formats (PNG, JPG).
2.	The system shall extract text content from uploaded PDF files using PyMuPDF.
3.	The system shall extract text content from uploaded DOCX files using python-docx.
4.	The system shall extract text content from uploaded PPTX files using python-pptx.
5.	The system shall extract embedded images from PDF and DOCX files and store them separately with a reference to the parent document.
6.	For scanned/image-only PDF pages, the system shall apply OCR (Tesseract) to extract text.
7.	The system shall split extracted text into chunks of approximately 800 tokens with 100-token overlap between consecutive chunks.
8.	The system shall generate a vector embedding for each text chunk using the configured embedding model.
9.	The system shall store embeddings in pgvector for semantic search.
10.	The system shall store document metadata (title, industry, client, technologies, upload date, file paths) in PostgreSQL.
11.	The system shall allow uploading multiple files as a single case study (e.g., PDF + images + drawings).
12.	Document ingestion shall run as a background task and not block the user interface.
13.	The system shall allow bulk upload of an entire folder of case studies.

2.2.2 Search & Retrieval

ID	Requirement
1.	The system shall accept free-text project descriptions as search input (minimum 20 characters, maximum 5000 characters).
2.	The system shall convert the search query into a vector embedding and perform cosine similarity search against all case study chunks.
3.	The system shall return the top N most relevant case studies (default N=10, configurable).
4.	Each result shall include: case study title, relevance/similarity score (0–100%), summary of why it's relevant, and industry/technology tags.
5.	Each result shall provide download links for all files associated with that case study (PDF, images, drawings).
6.	The system shall display extracted images from each case study as thumbnail previews in the results.
7.	The system shall use the LLM to generate a brief explanation of why each case study matches the query.
8.	The system shall save search history (query, results, timestamp) for each user.
9.	Results shall be returned within 5 seconds for a corpus of up to 500 case studies.

2.3 Tender Module

2.3.1 Tender Upload & Parsing

ID	Requirement
1.	The system shall accept government tender uploads in PDF format.
2.	The system shall also accept tender uploads in DOCX format.
3.	The system shall extract text from each page of the tender, preserving page numbers.
4.	The system shall extract tables from the tender and convert them to structured text.
5.	For scanned pages, the system shall apply OCR to extract text.
6.	The system shall split the tender into page-aware chunks of approximately 2000 tokens with 200-token overlap. Each chunk shall carry its source page number(s).

2.3.2 Requirement Extraction

ID	Requirement
1.	The system shall analyze each chunk using the LLM to extract individual requirements.
2.	Each extracted requirement shall include: description, category, page number, mandatory/optional flag, any associated deadline, and whether a document attachment is needed.
3.	Requirement categories shall include but not be limited to: certification, financial, legal, technical, experience, eligibility, submission format, evaluation criteria.
4.	The system shall merge and deduplicate requirements extracted from overlapping chunks.
5.	The system shall process the entire tender — no page or section shall be skipped regardless of document length.
6.	If a chunk fails to process (LLM error), the system shall retry up to 3 times before marking that chunk as “needs manual review” and continuing with the remaining chunks.

2.3.3 Report Generation

ID	Requirement
1.	The system shall generate a summary report in DOCX format containing all extracted requirements.
2.	The report shall include a requirements table with columns: Requirement, Category, Page Number, Mandatory (Yes/No), Attachment Needed (Yes/No), Status.
3.	The report shall include an executive summary section.
4.	The report shall include a document attachment checklist listing all documents that need to be submitted.
5.	The report shall list any requirements that could not be matched to company documents, flagged as “Missing”.
6.	The system shall also generate a PDF version of the report.

2.3.4 Document Matching

ID	Requirement
1.	The system shall maintain an indexed catalog of company documents (certificates, financials, legal docs, HR docs, technical docs).
2.	Company documents shall be tagged with: type/category, name, aliases/keywords, and file path.
3.	For each requirement that needs a document attachment, the system shall search the company document catalog using semantic similarity + keyword matching.
4.	The system shall assign a confidence score to each match: ≥0.85 = auto-match, 0.50–0.84 = suggested match, <0.50 = no match found.
5.	The system shall present all matches (auto and suggested) to the user for review before finalizing.
6.	The user shall be able to accept, reject, or manually assign a different document for each requirement.

2.3.5 Folder Creation

ID	Requirement
1.	Upon user confirmation, the system shall create a new folder named {TenderName}_{Date}.
2.	The folder shall contain: the original tender PDF, the generated summary report, and all confirmed/matched company documents.
3.	The folder shall organize company documents into a subfolder called “Required Documents”.
4.	The system shall make the complete folder available for download as a ZIP archive.

2.3.6 Progress Tracking

ID	Requirement
1.	The system shall provide real-time progress updates during tender analysis via WebSocket.
2.	Progress updates shall include: current step number, total steps, step label, and percentage complete.
3.	The analysis pipeline shall have 8 defined steps: Parse, Chunk, Extract, Merge, Match, Report, Folder, Review.
4.	If the user closes the browser during analysis, they shall be able to return and see the current progress or completed results.

2.4 Methodology Module

ID	Requirement
1.	The system shall accept methodology document uploads in PDF and DOCX format.
2.	The system shall index methodology documents with full-text search (PostgreSQL tsvector) and vector embeddings.
3.	The system shall allow users to search for methodology documents by name or topic.
4.	Search results shall appear within 2 seconds.
5.	The system shall return matching documents with: title, file type, relevance score, and upload date.
6.	The system shall allow users to download methodology documents directly from the search results.
7.	The system shall provide in-browser preview for PDF methodology documents.
8.	The system shall allow users to browse all methodology documents in a paginated list.
9.	The system shall support semantic search — e.g., searching "cybersecurity assessment" shall match a document titled "Network Security Methodology".

2.5 Dashboard

ID	Requirement
1.	The dashboard shall display three distinct clickable module cards: Case Study, Tender, and Methodology.
2.	Each card shall show the module name, icon, brief description, and count of documents in that module.
3.	The dashboard shall display a "Recent Activity" section showing the last 10 actions across all modules.
4.	The dashboard shall show summary statistics: total case studies, total tenders analyzed, total methodology docs.

3.6 File Management

ID	Requirement
1.	The system shall support file uploads up to 100 MB per file.
2.	The system shall store uploaded files in a structured directory organized by module and document type.
3.	The system shall generate unique file names to prevent collisions.
4.	The system shall support downloading any stored document via a secure, authenticated URL.
5.	The system shall support in-browser PDF preview without requiring download.
6.	The system shall validate file types on upload and reject unsupported formats with a clear error message.